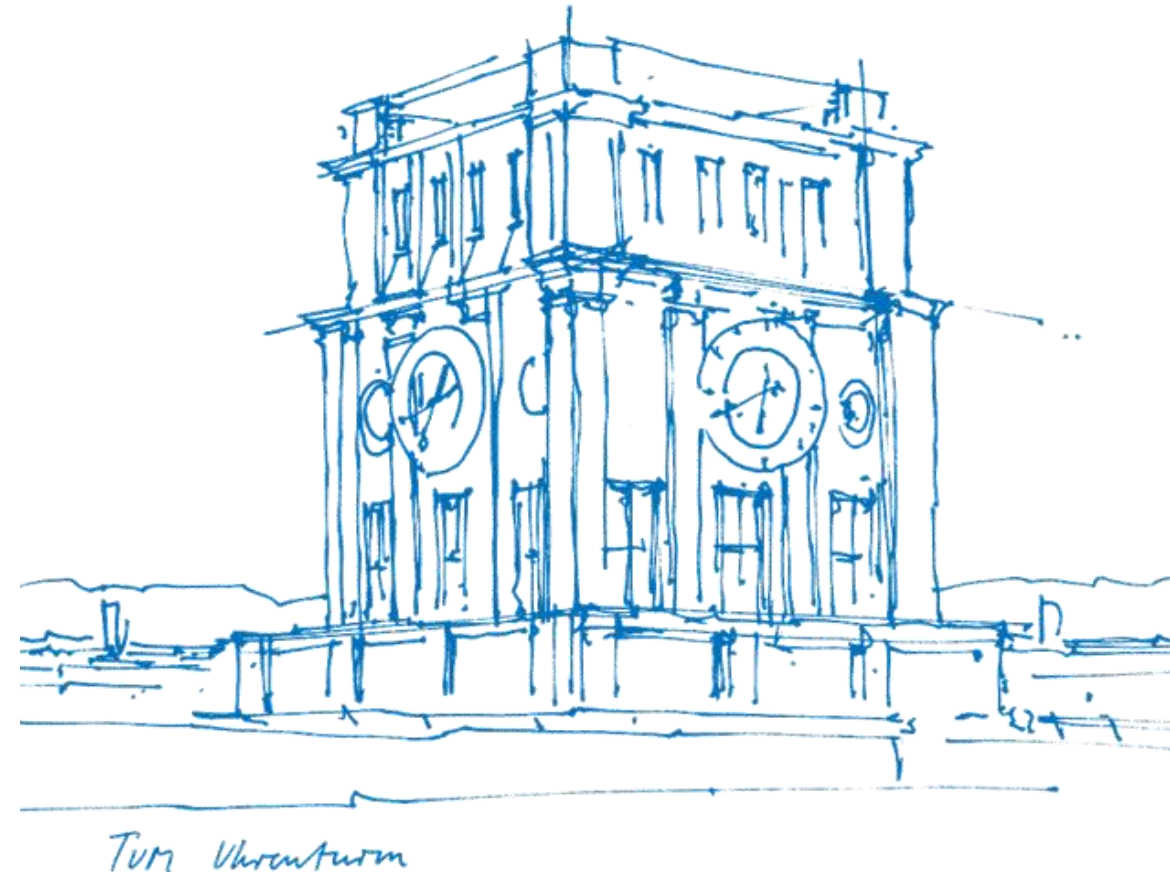


Tutorial 01

Set Theory & σ -Algebras

Xujie Zhao

29. April & 2. May 2025



- **Lecture:** 2.5h per week at 16:00-18:30 Wed, held by Prof. Niki Kilbertus in room MW2001
- **Tutorial:** In-Person, 2h by two time periods per week
 - 12:00-14:00 Tue, 03.11.018 (Group B33)
 - 10:00-12:00 Fri, 01.06.011 (Group E24)
- The slides are not guaranteed to be accurate! (Written by myself)
- **Language during the tutorial:** Mainly English, Deutsch may be used if needed, Chinese might be used during calculation (that would be an exception and will be repeated in English)
- **Exam:** August 1, 2025 from 08:00-10:00, Cheatsheet TBD
- **Materials of this tutorial:** Please scan the QR-Code 



Overview (referring to last year's lecture)

- **Discrete Probability Spaces**

- Probability Space, Event, Random Variable
- Special Distributions
- Markov's and Chebyshev's Inequalities

- **Continuous Probability Spaces**

- Normal Distribution, Exponential Distribution
- Central Limit Theorem

- **Statistics**

- Estimators
- Confidence Intervals
- Hypothesis Testing

- **Stochastic Processes**

- Markov Chains
- Queues

△ Derivative and Integral used!
(Analysis für Informatiker MA0902)

△ Matrix used!
(Lineare Algebra MA0901)

Personally advices...

- PART I. Recap of the Lecture
- PART II. Exercises from the Exercise Sheet
- PART III. Additional Exercises (if time permits)

I. Recap

(From start of the script to the Proposition 1.17)

- **Set Theory**

- Basic Definitions
- De Morgan's Laws
- Samples (sample space), Events, Event Systems

- **σ -Algebra**

- Definition
- Intersection of σ -algebras
- Generator of a σ -algebra

- **Measurable space and sets**

- Definition
- Induced σ -algebra

- **Measure and measure space**

- Definition
- Properties of measures

II. Exercise Sheet



Exercise 1. Let $\Omega \neq \emptyset$ and let $\{A_\alpha \in \mathcal{P}(\Omega) \mid \alpha \in I\}$ be an arbitrary set of subsets of Ω , I the indexing set. Then the following two identities hold (commonly referred to as *de Morgan's law*):

$$\left(\bigcap_{\alpha} A_{\alpha}\right)^c = \bigcup_{\alpha} A_{\alpha}^c, \quad \left(\bigcup_{\alpha} A_{\alpha}\right)^c = \bigcap_{\alpha} A_{\alpha}^c$$

II. Exercise Sheet



Exercise 2. Let Ω_1, Ω_2 be non-empty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and $\mathcal{C} \subset \mathcal{P}(\Omega_2)$ an arbitrary collection of subsets of Ω_2 . Then the following statements hold.

(i) If $A, B \subset \Omega_2$ and $A \subset B$, then $f^{-1}(A) \subset f^{-1}(B)$.

(ii) The inverse image of the union is equal to the union of the inverse images, meaning

$$f^{-1}\left(\bigcup_{A \in \mathcal{C}} A\right) = \bigcup_{A \in \mathcal{C}} f^{-1}(A) .$$

(iii) Given a subset $A \subset \Omega_2$, the inverse-image of the complement is equal to the complement of the inverse image, meaning

$$f^{-1}(A^c) = (f^{-1}(A))^c$$

II. Exercise Sheet



Exercise 3. Suppose $\mathcal{A}$ is a σ -algebra over $\Omega \neq \emptyset$, let $A_0, A_1, A_2, \dots$ be a countable collection of sets in $\mathcal{A}$ and let $m \in \mathbb{N}$. Then the sets

$$\Omega, \quad A_0 \setminus A_1, \quad \bigcup_{i=0}^m A_i, \quad \bigcap_{i=0}^m A_i, \quad \bigcap_{i \in \mathbb{N}} A_i$$

are also in $\mathcal{A}$.

[Hint: Use de Morgan's law for the intersection properties.]

II. Exercise Sheet



Exercise 4. True or false? Justify your answers either by a proof or a counterexample.

- (i) The set $\mathcal{A} = \{\emptyset, \{1, 2\}, \{1, 4\}, \{2, 3\}, \{3, 4\}, \{1, 2, 3, 4\}\} \subset \mathcal{P}(\Omega)$ is a σ -algebra over $\Omega = \{1, 2, 3, 4\}$.
- (ii) If $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{P}(\mathcal{A})$ are σ -algebras over Ω , then so is $\mathcal{A}_1 \cup \mathcal{A}_2$.
- (iii) The set $\mathcal{A} := \{A \subset \Omega \mid A \text{ or } A^c \text{ is countable}\}$ is a σ -algebra over Ω .
- (iv) The set $\mathcal{A} := \{A \subset \Omega \mid A \text{ or } A^c \text{ is finite}\}$ is a σ -algebra over Ω if and only if Ω is finite.

[Hint: You can use the following facts without further justification: The countable union of countable sets is again countable. Subsets of countable sets are again countable.]

II. Exercise Sheet



Exercise 5. Let $\Omega \neq \emptyset$ be a set, I be a nonempty index set (finite, countable or even uncountable) and let $\mathcal{A}_\alpha$ be a σ -algebra over Ω for each $\alpha \in I$ in the index set. Then $\mathcal{A} := \bigcap_{\alpha \in I} \mathcal{A}_\alpha$, the intersection of all the sigma-algebras is also a σ -algebra over Ω .

Remark 1. Let Ω be a nonempty set and let $\mathcal{S} \subset \mathcal{P}(\Omega)$ also be nonempty. Then by Exercise 5, we have that

$$\mathcal{A}_\sigma(\mathcal{S}) := \bigcap \{ \mathcal{A} \subset \mathcal{P}(\Omega) \mid \mathcal{A} \subset \mathcal{S}, \mathcal{A} \text{ is a } \sigma\text{-algebra over } \Omega \}$$

is well-defined, a σ -algebra over Ω and by definition the smallest σ -algebra over Ω containing $\mathcal{S}$. We call $\mathcal{A}_\sigma(\mathcal{S})$ the σ -algebra generated by $\mathcal{S}$.

II. Exercise Sheet



Exercise 6. Let Ω_1, Ω_2 be nonempty sets, $f : \Omega_1 \rightarrow \Omega_2$ an arbitrary mapping and let $\mathcal{A}, \mathcal{B}$ be σ -algebras over Ω_1 and Ω_2 respectively. Then the *inverse image of $\mathcal{B}$ under f* and the *image or push-forward of $\mathcal{A}$ under f* defined as

$$f^{-1}(\mathcal{B}) := \{f^{-1}(B) \mid B \in \mathcal{B}\} \quad \text{and} \quad f_*(\mathcal{A}) := \{B \subset \Omega_2 \mid f^{-1}(B) \in \mathcal{A}\}$$

are σ -algebras over Ω_1 and Ω_2 respectively.

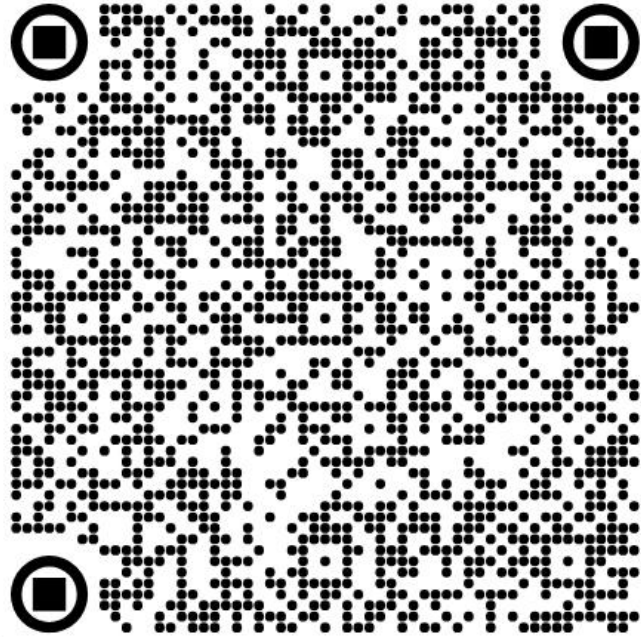
[Hint: Use statements from the second exercise.]

III. Additional Exercises

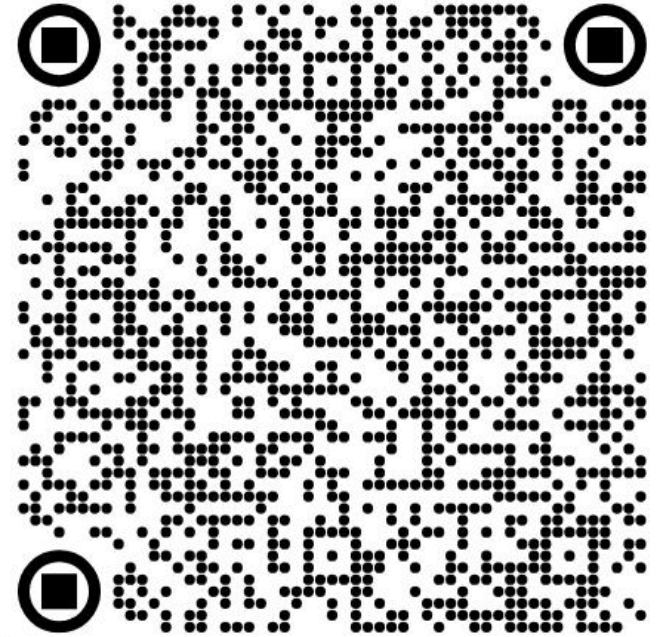
Exercise 7. Anna flips a fair coin countable but infinitely many times. The set Ω consists of all infinite sequences of the outcomes "Heads" and "Tails".

- (a) Show that the set Ω is uncountable.
- (b) Let $\mathcal{A}$ be the set of all subsets $A \subseteq \Omega$ such that either A and $\Omega \setminus A$ is countable. Prove that $\mathcal{A}$ is a σ -Algebra.

Feedback & Materials



https://docs.google.com/forms/d/e/1FAIpQLSfa6RXGN59bmV6DvR_rmnZorPO50nV-LEOjJm6fMwqYciPfrw/viewform?usp=dialog



<https://drive.google.com/drive/folders1rBeL0NKHuYzrULK3Fr4CGr8svRhyTNOO?usp=sharing>